

# Contrasting responses to aridity by different-sized decomposers cause similar decomposition rates across a precipitation gradient

Reviewed Preprint

v2 • August 20, 2024

Revised by authors

Reviewed Preprint

v1 • February 26, 2024

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## Abstract

Litter decomposition is expected to be positively associated with precipitation despite evidence that decomposers of varying sizes have different moisture dependencies. We hypothesized that higher tolerance of macro-decomposers to aridity may counterbalance the effect of smaller decomposers, leading to similar decomposition rates across climatic gradients. We tested this hypothesis by placing plant litter baskets of different mesh sizes in seven sites along a sharp precipitation gradient, and by characterizing the macro-decomposer assemblages using pitfall trapping. We found that decomposers responded differently to precipitation levels based on their size. Microbial decomposition increased with precipitation in the winter while macro-decomposition peaked in arid sites during the summer. This led to similar overall decomposition rates across the gradient except in hyper-arid sites. Macro-decomposer richness, abundance and biomass peaked in arid environments. Our findings highlight the importance of macro-decomposition in arid-lands, possibly resolving the dryland decomposition conundrum, and emphasizing the need to contemplate decomposer size when investigating zoogeochemical processes.

### eLife Assessment:

This **fundamental** study substantially advances our understanding of the role of different-sized soil invertebrates in shaping the rates of leaf litter decomposition, using an experiment across seasons along an aridity gradient. The authors provide **compelling** evidence that the summed effects of all invertebrates (with large-sized invertebrates being more active in summer and small-sized invertebrates in winter) on decomposition rates result in similar levels of leaf litter decomposition across seasons. The work will be of broad relevance to ecosystem ecologists interested in soil food webs, and researchers interested in modeling carbon cycles to understand global warming.

<https://doi.org/10.7554/eLife.93656.2.sa2>

## Introduction

Litter decomposition is a key process determining elemental cycling in terrestrial ecosystems<sup>1</sup>. Decomposition is controlled by climate, litter quality and origin, and the identity and abundance of microbial and faunal decomposers<sup>2–4</sup>. Climate regulates decomposition rates directly, but also indirectly by influencing food-web structure and dynamics<sup>5</sup>. Thus, understanding how climate and decomposers interact is a key step in explaining variation in plant litter decomposition across ecosystems and seasons, and in forecasting the consequences of climate change and biodiversity loss on elemental cycling.

Theory suggests that decomposition is positively associated with moisture and temperature<sup>4</sup>. Cross-site studies, reviews, and meta-analyses verified this global pattern, showing that plant litter decomposition in microbial litter bags is indeed faster under warm and wet conditions than under cool and dry conditions<sup>6–11</sup>. This well accepted realization implicitly assumes that microorganisms dominate plant litter decomposition, largely ignoring the growing recognition that animals may play an important role in litter cycling. This role includes mineralizing and excreting assimilated plant nutrients, fragmenting and partly decomposing plant material, and transporting detritus to microbial havens<sup>12–19</sup>.

Attempts to explore how climate affects faunal decomposition revealed a similar positive association with temperature and precipitation<sup>20,21</sup>. This global pattern, however, may be confounded by compiling the effect of all decomposer fauna together, ignoring the well-established understanding that soil animals of various size groups respond differently to climate<sup>4,22</sup>. Specifically, larger arthropods can survive and remain active during hot and dry periods when smaller organisms cannot<sup>23</sup>. Indeed, handful evidence shows that macro-detritivorous arthropods dominate litter and wood decomposition in warm drylands, especially during warm and dry seasons<sup>24–26</sup>. This suggests that the conceptualization of how animals and climate interact to regulate decomposition rates requires considering the effects of meso-decomposers and those of macro-decomposers separately, particularly in warm drylands.

Detritivorous animals are expected to be exceptionally abundant in arid ecosystems where plant detritus is prevalent year around but green plant material is available predominantly in short pulses following precipitation events<sup>27</sup>. Macrofauna are physiologically and morphologically more adapted to aridity than mesofauna<sup>23</sup>. Moreover, their large size enables them to remain active during long dry periods by shuttling between existing and self-engineered climatic havens and the hostile foraging grounds on the surface<sup>28</sup>. Consequently, macro-decomposition should be especially important in hot moisture-deprived habitats and periods, whereas the activity of microbes and mesofauna is expected to be minimal.

The predicted negative association between moisture and macro-decomposition in drylands may be reversed in hyper-arid environments. In these environments, the extreme climatic conditions and scarce and unpredictable plant litter availability may limit macro-decomposer populations, diminishing macro-decomposition rates with increasing aridity. Consequently, and in sharp contrast to smaller organisms, macro-decomposition should follow a hump-shaped response to precipitation that peaks in arid ecosystems.

To test this novel hypothesis, we examined the climate dependency of plant litter decomposition by microorganisms, meso-decomposers and macro-decomposers along a sharp aridity gradient spanning from mean annual precipitation of only 22 mm to 526 mm. This gradient represents hyper-arid, arid, semiarid and dry sub-humid Mediterranean climates. We repeated the experiment during hot summer with no precipitation and again during cooler and wetter winter. We hypothesized that both microbial and mesofaunal decomposition should increase with

increasing precipitation during the winter, but during the dry summer contribute only minimally to plant litter decomposition across the aridity gradient. In the dry summer, macrofaunal decomposition should follow a hump-shaped response to precipitation, increasing from hyper-arid to arid sites and decreasing gradually in more mesic semi-arid and Mediterranean sites. We also predicted that the opposing climatic dependencies of macrofauna and microorganisms and mesofauna should lead to similar overall decomposition rates across the precipitation gradient except in the hyper-arid sites in which decomposers activity is predicted to be minimal regardless of organism size (**Fig. 1**). To reveal the mechanism, we sampled macro-decomposers across the aridity gradient and the two seasons, using pitfall traps. We predicted hump-shaped relationships between precipitation and the abundance, richness, and biomass of macro-decomposers that peak in arid ecosystems.

## Methods

We performed a manipulative litter mass loss experiment across seven sites representing a sharp mean annual precipitation (MAP) gradient ranging from hyper-arid desert to Mediterranean maquis (**Fig. 2A**, **Table 1**). All sites were chosen to be on calcareous soils formed upon sedimentary limestone rock in natural habitats. The mean annual temperature varies only slightly across sites from 18.7°C to 22.3 °C. The exact study sites were determined to ensure minimal human disturbance during the year-long experiment. In each of the seven sites we installed litter baskets of three different mesh sizes that control organismal access to litter: Micro-baskets allowing entry of only microorganisms (<200 µm), meso-baskets allowing entry of microorganisms and mesofauna (<2 mm), and macro-baskets that were identical to the meso-baskets but with side openings that allow entry of macrofauna (<2 cm). Litter baskets were filled with leaf litter belonging to the annual grass *Stipa capensis* Thunb. that is native to all seven study sites. This approach allowed us to compare decomposition rates across sites without using exogenous litter that may decompose in a very different rate than local litter<sup>3</sup>. Twenty-five blocks, each including the three basket types (**Fig. 2B**), were installed in each site for two consecutive experimental periods - a wet cool winter and a dry hot summer (2 periods x 7 sites x 3 treatments x 25 blocks = 1050 baskets in total). We also characterized the macro-decomposer assemblage in each site during the two seasons using pitfall trapping.

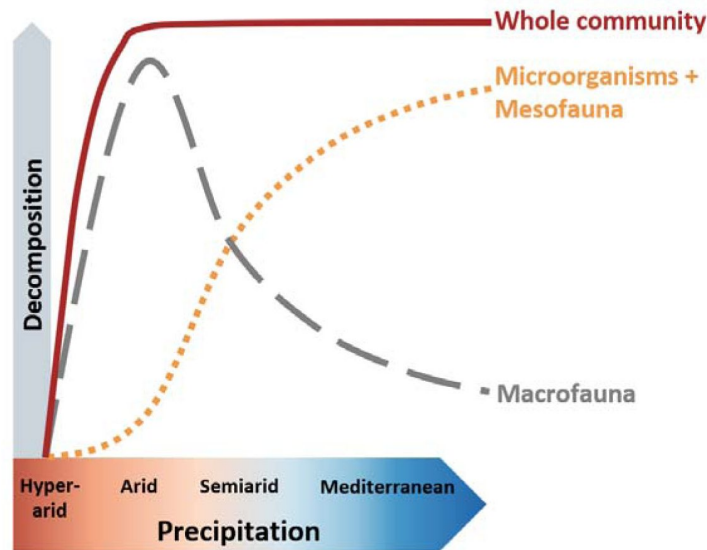
### Litter basket experiment

We collected *S. capensis* litter from the Avdat site in the summer of 2020 and air dried it. We sorted the litter to remove litter belonging to any other species and assigned  $3 \pm 0.0001$  g (Mettler Toledo MS105DU) to each litter basket. Thirty additional litter samples were oven dried at 60 °C for 48 hours and weighed again for determination of initial moisture content. The 14 x 13 x 3.6 cm litter baskets were prepared of a 12 mm mesh galvanized welded metal, lined at the bottom with a 1.5 mm fiberglass mesh to prevent litter loss, and covered from all sides (including top and bottom) with a 2 mm metal mesh to exclude termites. In the macro-baskets, three 2 x 2 cm windows were cut at each of the four sides. The windows were cut approximately 1 cm above ground level to allow macrofaunal access but prevent accidental litter spill. This may slightly reduce macrofaunal access, making our estimations of the macrofaunal effect conservative. In the micro-baskets, we placed the litter within a polyethylene 200 µm mesh bag. In the macro- and meso-baskets, we laid a 2 cm heavy metal mesh over the litter to minimize litter loss due to wind.

We installed the first batch of 525 litter baskets in the field in November-December 2020. All blocks were placed around similarly sized bushes of locally distributed species and tethered to the ground using metal stakes. We collected the baskets in May-July 2021 and replaced them with a new similar batch that was later collected in October-November 2021. At the end of each season, the collected baskets were transported to the laboratory in sealed Ziplock bags. Any litter spilled

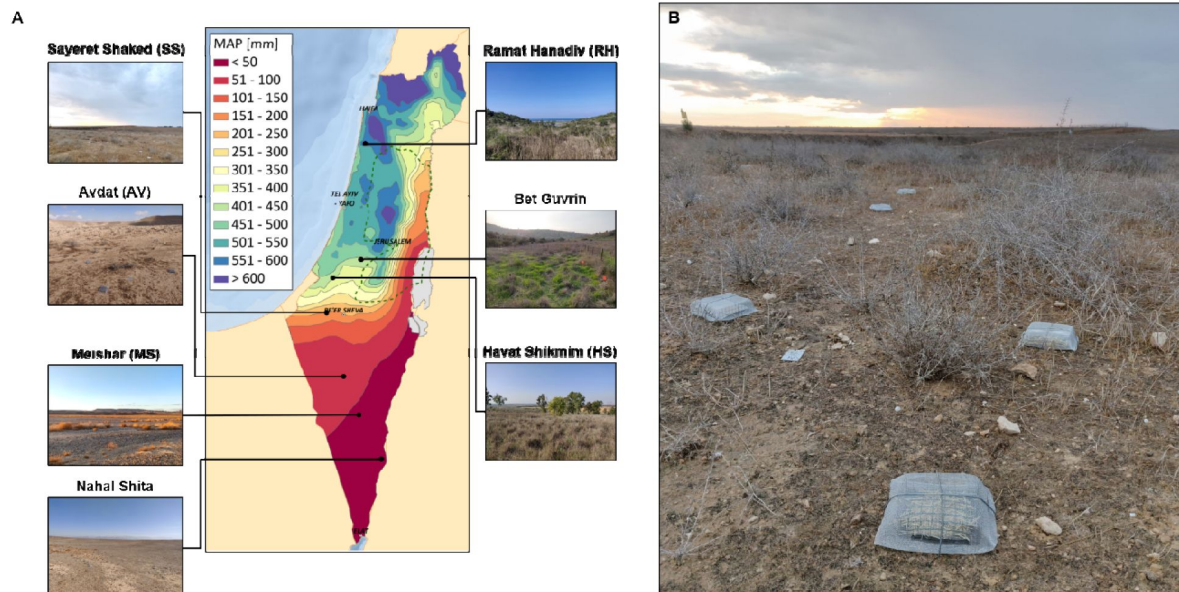
**Figure 1 -**

Hypothetical climate dependence of litter decomposition by microorganisms and mesofauna (dotted orange curve), by macrofauna (grey - dash), and by the whole decomposer community (maroon - solid).



**Figure 2 -**

(A) Locations and landscapes of the seven experimental sites across a precipitation gradient from 22 to 526 mm mean annual precipitation (MAP). (B) A block of three litter baskets in the Sayeret Shaked site. Macro-basket in front, meso-basket on the right and micro-basket on the left. The precipitation map courtesy of the Hebrew University GIS center.



| Site              | Abb. | Coordinates                  | MAT*<br>[°] | MAP*<br>[mm] | AI <sub>U</sub> *<br>(MAP/PET) | Climate                       | Winter<br>experiment      | Summer<br>experiment |
|-------------------|------|------------------------------|-------------|--------------|--------------------------------|-------------------------------|---------------------------|----------------------|
| Ramat<br>Hanadiv  | RH   | 32°33'22.4"N<br>34°56'26.6"E | 20.2        | 526          | 0.518                          | Dry subhumid<br>Mediterranean | 3.12.2020 –<br>27.6.2021  | 27.6 –<br>27.10.2021 |
| Bet<br>Guvrin     | BG   | 31°35'54.7"N<br>34°54'14.2"E | 20.9        | 403          | 0.370                          | Semiarid                      | 2.12.2020 –<br>13.6.2021  | 13.6 –<br>21.10.2021 |
| Havat<br>Shikmim  | HS   | 31°30'49.7"N<br>34°41'18.8"E | 19.8        | 367          | 0.364                          | Semiarid                      | 2.12.2020 –<br>13.6.2021  | 13.6 –<br>21.10.2021 |
| Sayeret<br>Shaked | SS   | 31°16'05.7"N<br>34°39'12.9"E | 20.0        | 148          | 0.145                          | Arid                          | 26.11.2020<br>– 23.5.2021 | 23.5 –<br>21.10.2021 |
| Avdat             | AV   | 30°47'02.3"N<br>34°46'13.3"E | 18.7        | 84           | 0.089                          | Arid                          | 26.11.2020<br>– 23.5.2021 | 23.5 –<br>21.10.2021 |
| Meishar           | MS   | 30°27'04.2"N<br>34°56'03.0"E | 20.8        | 33           | 0.029                          | Hyperarid                     | 10.12.2020<br>– 12.7.2021 | 12.7 –<br>8.11.2021  |
| Nahal<br>Shita    | NS   | 30°08'29.4"N<br>35°07'36.6"E | 22.3        | 22           | 0.017                          | Hyperarid                     | 10.12.2020<br>– 12.7.2021 | 12.7 –<br>8.11.2021  |

\* Climatic data extracted from [http://www.meteo-tech.co.il/hanadiv\\_new/hanadiv\\_en.asp](http://www.meteo-tech.co.il/hanadiv_new/hanadiv_en.asp) (RH), courtesy of Shaily Dor-Haim (SS), and extracted from <https://ims.gov.il/en> (all other sites).

**Table 1 -**

### **Properties of the seven experimental sites**

during transportation was weighed and the weight loss was incorporated in the calculations. Leaf litter in each basket was first screened for adulteration from leaf litter of other species, following which the *S. capensis* litter was oven dried at 60 °C for 48 hours and weighed. To account for dust accumulation on the litter we applied an ash correction procedure<sup>29</sup>. We burned and weighed five sub-samples from each site-season-treatment combination (550 °C for 5 hours) and calculated the combination-specific mean ash content. The final litter mass was corrected for ash content based on these calculations. We burned and weighed 15 additional samples of *S. capensis* litter that were not placed in the field and calculated the mean ash content of the initial litter. The initial litter mass was corrected accordingly. The rate of litter removal from each basket was calculated as the difference between the ash corrected final dry litter mass and the ash and moisture corrected initial litter mass, divided by the number of days the litter spent in the field.

## Pitfall trapping

We characterized the macro-decomposer assemblages by setting up 20 pitfall traps for 5-7 days at each site during each experimental period. Wet season traps were opened in February 2021, whereas the dry season traps were opened between late August and early October. We installed traps by placing two 10 cm diameter X 7.5 cm deep plastic containers one inside another such that the opening was flushed with the ground. We added to each trap 150 ml of preservative, which comprised of 40% absolute ethanol, 40% distilled water and 20% Propylene glycol. Traps were covered with steel mesh of large mesh size to prevent small mammals and reptiles from falling inside. At the end of the 5-7 days, samples were collected and transferred to 70% ethanol. Samples were sorted and identified to morphospecies level in the lab. Only animals larger than 2 mm were included in the analysis. Sub-samples were freeze-dried and weighed (Mettler Toledo MS105DU) for biomass estimation of each morphospecies.

## Analytical procedures

We first fitted a linear mixed model (LMM) to the litter removal rate data, including experimental site, experimental season, mesh size and all interaction terms as fixed effects. The random effect of the experimental spatial blocks was found insignificant using a simulation test with 9999 simulations (RLRT=2.1, P-value=0.07). Therefore, we assessed the effects of the site, season and mesh size on the litter removal rate using a full factorial analysis of variance, followed by Tukey HSD pairwise comparisons. We calculated the contribution of each size group to litter mass loss by block. Microbial contribution was defined as the mass loss from micro-baskets; mesofaunal contribution was calculated as the difference between mass loss from meso- and micro-baskets; macrofaunal contribution was calculated as the difference between mass loss from macro- and meso-baskets; whole-community decomposition was defined as the mass loss from macro-baskets. We modeled the relationship between MAP and each of these contributions using Locally Estimated Scatterplot Smoothing (LOESS). We used the same method to model the relationship between MAP and the macro-decomposer abundance, biomass and morphospecies richness in each season. We assessed differences in the macro-decomposer assemblage among experimental sites and seasons using a principal coordinates analysis (PCoA) with individual traps as the sampling units and Bray-Curtis index (BC) as the dissimilarity metric. We tested for differences across sites and seasons in macro-decomposer assemblage using a permutational multivariate analysis of variance (PERMANOVA), followed by pairwise comparisons between sites using the Benjamini-Hochberg P-value adjustment. BC indices between site-season combinations were calculated as well, based on the summed abundances across traps. To explore which macro-decomposer groups dominate the different sites and seasons, we classified the identified morphospecies to ten macro-decomposer taxa: Archaeognatha (bristletails), Coleoptera (beetles), Diplopoda (millipedes), Formicidae (ants), Gastropoda (snails and slugs), Grylloidea (crickets), Isoptera (termites), Lumbricina (earthworms), Oniscidea (woodlice) and Zygentoma. Then we summed the abundance, richness and biomass from each group in each trap. We used the



abundance data to fit the group scores onto the PCoA ordination. Litter removal data was analyzed using the ‘stats’, ‘lme4’<sup>30</sup> and ‘RLRsim’<sup>31</sup> packages from R software (version 4.3.0)<sup>32</sup>, whereas assemblage data was analyzed using the ‘vegan’ package<sup>33</sup>.

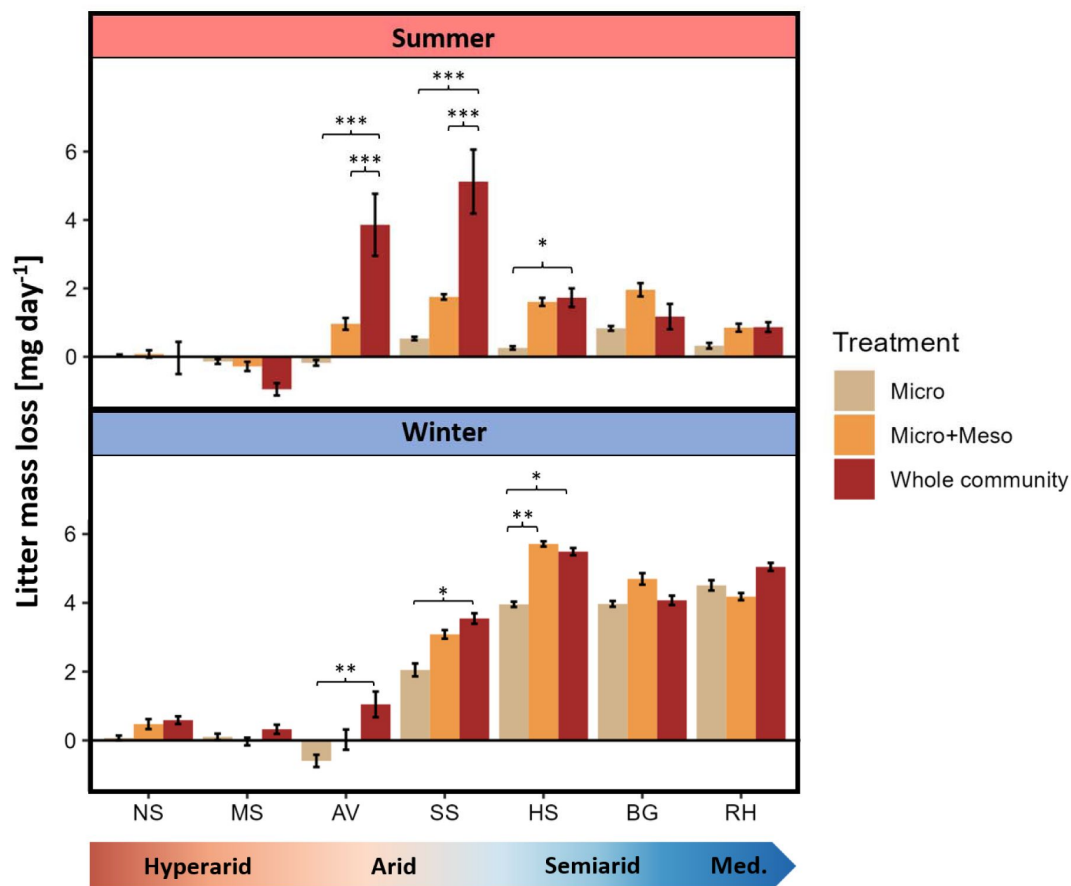
## Results

Litter removal rate differed across seasons, sites and mesh sizes, and all interactions between these factors were found significant as well (**Table 2**). On average, the litter removal rate was 2.6-fold higher in winter than in summer, 1.6-fold higher in meso- than in micro-baskets and 1.3-fold higher in macro- than in meso-baskets. Litter removal was negligible in the hyperarid sites during both seasons, while it was highest in the arid sites during summer and in the more mesic sites during winter (**Fig. 3**). Within site and season comparisons between mesh sizes yielded significant differences only in Avdat, Sayeret Shaked and Havat Shikmim, indicating that faunal effects on decomposition were found only under arid to dry-semiarid conditions (**Fig. 3**). Both macro- and mesofaunal effects were detected in the arid sites (Avdat, Sayeret Shaked), whereas the semiarid Havat Shikmim site exhibited only a mesofaunal effect during both seasons (**Fig. 3**). The macrofaunal, mesofaunal and microorganismal contributions to litter mass loss peaked under arid, semiarid and Mediterranean climate conditions, respectively (**Fig. 4**). Whole-community litter removal rates were dictated by microorganisms in winter and by macrofauna in summer, resulting in comparable rates across the aridity gradient from Mediterranean to arid climate at the annual scale (**Fig. 4**). In total, the whole-community litter removal rate peaked in Sayeret Shaked (MAP=148 mm) and significantly decreased under drier and wetter conditions (**Fig. 4**).

Macro-decomposer abundance, biomass and morphospecies richness peaked in the arid sites during both seasons (**Fig. 5**). The macro-decomposer assemblage differed significantly across sites ( $F_{6,229}=10.6$ ,  $P\text{-value}<0.001$ ) and across seasons ( $F_{1,229}=13.1$ ,  $P\text{-value}<0.001$ ), where woodlice, millipedes and snails were substituted by ants and termites with increasing aridity (**Fig. 6A, B**). Assemblage was significantly affected by the interaction between site and season too ( $F_{6,229}=5.4$ ,  $P\text{-value}<0.001$ ). The experimental site explained much of the assemblage variability across traps ( $R^2=0.18$ ), whereas experimental season accounted for a smaller fraction ( $R^2=0.04$ ), and site-season interaction played an intermediate role ( $R^2=0.10$ ). All pairwise comparisons across sites yielded significant differences in assemblage (**Table S1**). In general, ants were the most abundant group, whereas beetles accounted for most of the biomass. However, under mesic conditions, woodlice (Ramat Hanadiv site), millipedes (Ramat Hanadiv and Bet Guvrin) and snails (Havat Shikmim) were dominant (**Fig. 6C**). The Ramat Hanadiv assemblage was distinctively different from all other sites (**Fig. 6A**), as demonstrated by very high Bray-Curtis dissimilarity indices compared to the other sites, regardless of the season (**Table S2**). There were parallels between the spatial and temporal axes of aridity, as winter communities in the most arid sites (Nahal Shita, Meishar and Avdat) were relatively similar to the summer communities of the more mesic sites (Sayeret Shaked, Havat Shikmim and Bet Guvrin) (**Fig. 6A, B**; **Table S2**). The arid sites, where macro-decomposer assemblages flourished and were responsible for the highest litter mass loss, showed interesting seasonal dynamics. Bray-Curtis dissimilarity across seasons was higher in Sayeret Shaked than in Avdat ( $BC=0.79$  and  $0.72$ , respectively). Cross-site dissimilarity between Sayeret Shaked and Avdat was higher in winter than in summer ( $BC=0.85$  and  $0.72$ , respectively).

## Discussion

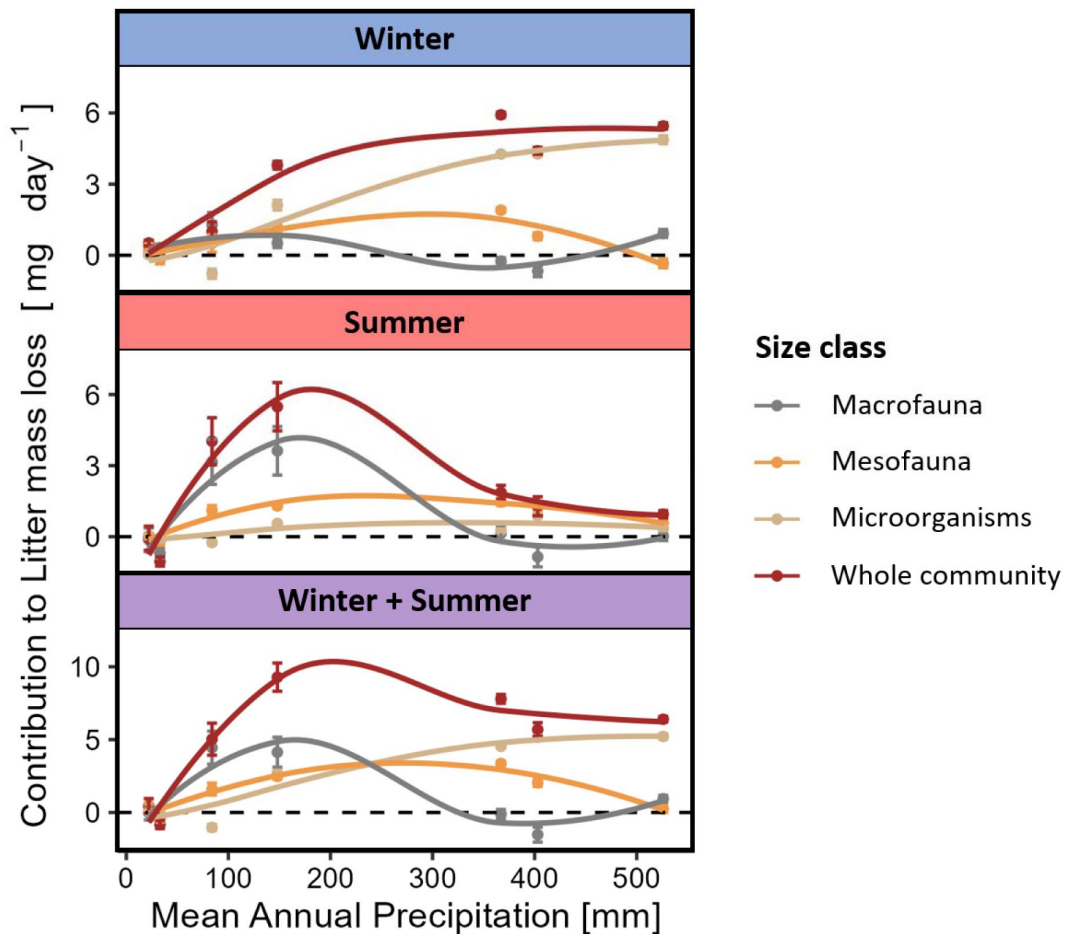
Our goal was to investigate how climate interacts with soil biota of different size categories to influence litter decomposition. We used litter baskets of different mesh sizes that were placed along a sharp precipitation gradient during hot-dry summer and again during colder-wetter winter. Our results suggest that decomposers respond differently to precipitation levels based on



**Figure 3 -**

Litter removal rate (mean  $\pm$  se) from baskets with different mesh sizes across sites and seasons. Asterisks represent significant differences between mesh sizes within site and season: \* P-value<0.05, \*\*P-value<0.01, \*\*\*P-value<0.001. NS – Nahal Shita; MS – Meishar; AV – Avdat; SS – Sayeret Shaked; HS – Havat Shikmim; BG – Bet Guvrin; RH – Ramat Hanadiv. Negative values may represent cases in which physical cleaning and ash correction failed to correct for all dust accumulation on the litter or cases in which exogenous litter may have penetrated the baskets. Our findings were not sensitive to these negative values.





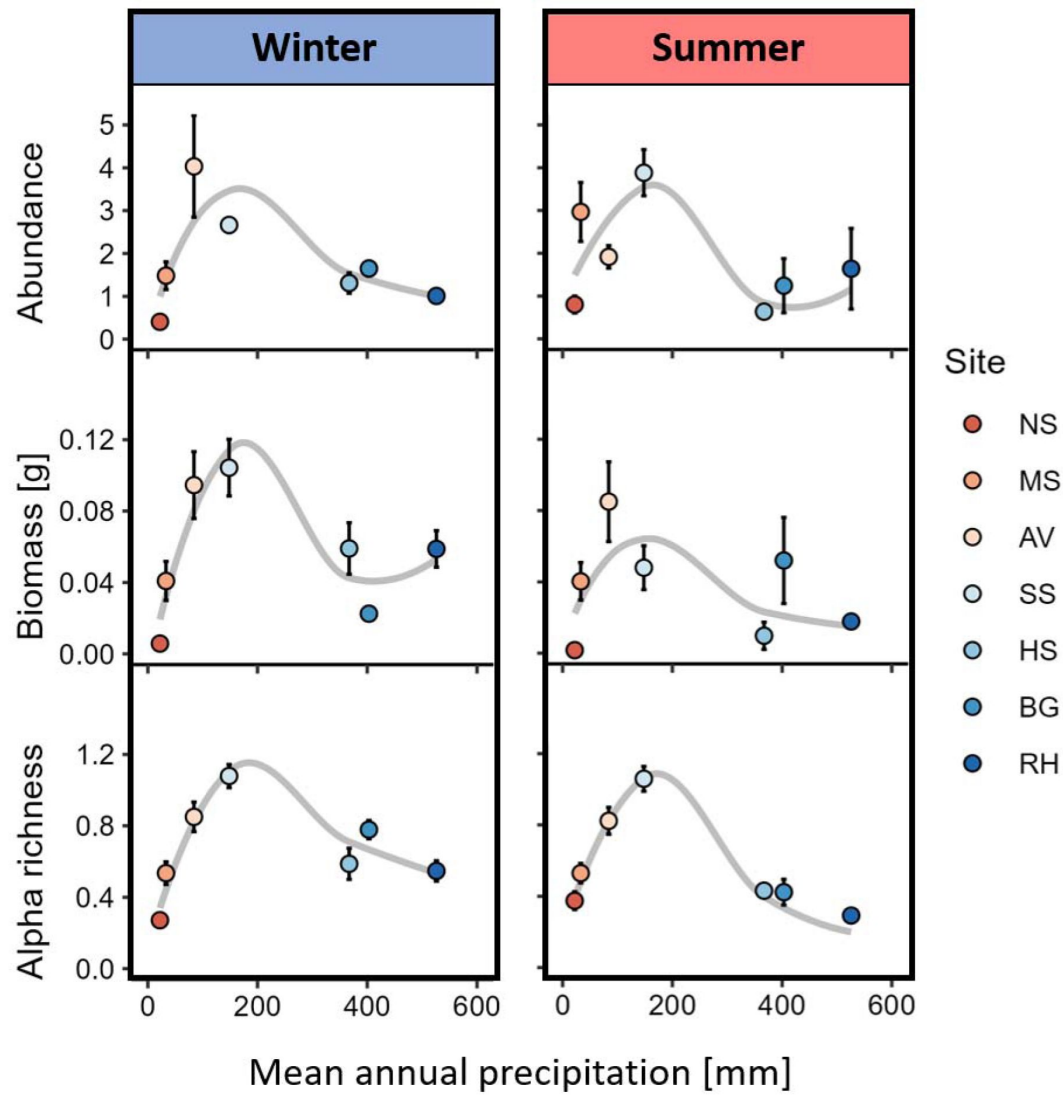
**Figure 4 -**

Contribution of different organism size classes to litter removal (mean  $\pm$  se) across the precipitation gradient during summer, winter and both seasons combined. Macrofaunal contribution was calculated as the within-block difference between macro- and meso-baskets; mesofaunal contribution as the difference between meso- and micro-baskets; microbial and whole community contributions represent litter removal rates in the micro- and macro-baskets, respectively. Curves were fitted to data using local estimation scatterplot smoothing (LOESS). Negative values may represent cases in which physical cleaning and ash correction failed to correct for all dust accumulation on the litter or cases in which exogenous litter may have penetrated the baskets. Our findings were not sensitive to these negative values.

|                                  | Df   | Sum Sq   | Mean Sq  | F value | Pr(>F) |
|----------------------------------|------|----------|----------|---------|--------|
| <b>Site</b>                      | 6    | 0.002092 | 0.000349 | 171.946 | <0.001 |
| <b>Season</b>                    | 1    | 0.000679 | 0.000679 | 334.94  | <0.001 |
| <b>Mesh size</b>                 | 2    | 0.000267 | 0.000133 | 65.77   | <0.001 |
| <b>Site : Season</b>             | 6    | 0.00112  | 0.000187 | 92.044  | <0.001 |
| <b>Site : Mesh size</b>          | 12   | 0.000392 | 3.27E-05 | 16.11   | <0.001 |
| <b>Season : Mesh size</b>        | 2    | 1.77E-05 | 8.9E-06  | 4.368   | 0.0129 |
| <b>Site : Season : Mesh size</b> | 12   | 0.000129 | 1.08E-05 | 5.306   | <0.001 |
| <b>Residuals</b>                 | 1008 | 0.002044 | 0.000002 |         |        |

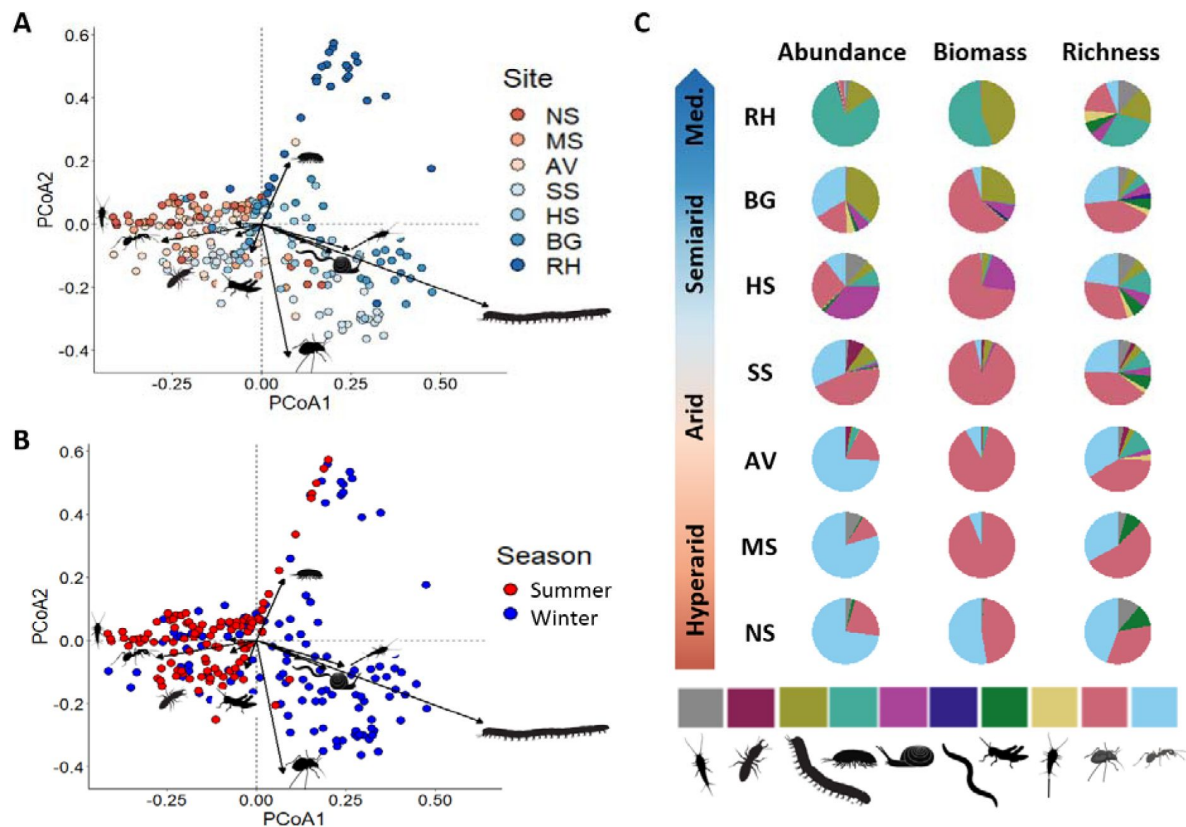
**Table 2 -**

**Results of a full-factorial analysis of variance in litter removal rate across mesh sizes, experimental sites and seasons.**



**Figure 5 -**

Macro-decomposer abundance, biomass and alpha morphospecies richness across the precipitation gradient in the two experimental seasons (mean  $\pm$  se). Values are averaged across traps and divided by the number of trapping days. NS – Nahal Shita; MS – Meishar; AV – Avdat; SS – Sayeret Shaked; HS – Havat Shikmim; BG – Bet Guvrin; RH – Ramat Hanadiv.



**Figure 6 -**

(A) and (B) Graphical representation of the first two axes of a principal coordinate analysis on the macro-decomposer assemblage data. Colors represent experimental sites in A and seasons in B. Arrows represent taxonomic group scores fitted onto the PCoA ordination. (C) Distribution of abundance, biomass and morphospecies richness among macro-decomposer taxonomic groups in each site across the aridity gradient. NS – Nahal Shita; MS – Meishar; AV – Avdat; SS – Sayeret Shaked; HS – Havat Shikmim; BG – Bet Guvrin; RH – Ramat Hanadiv. Color codes (left to right in panel C): grey – Zygentoma, burgundy – termites, olive green – millipedes, turquoise – woodlice, pink – snails and slugs, purple – earthworms, dark green – crickets, pale yellow – bristletails, red – beetles, pale blue – ants.

their size, leading to similar overall decomposition rates across the gradient, except in hyper-arid sites. We found that microbial decomposition was minimal during the summer. In the winter, microbial decomposition was positively associated with precipitation, governing the whole-community decomposition. In both seasons, mesofaunal decomposition was moderate and followed a hump-shaped response to precipitation, peaking in semiarid sites. Macro-decomposition contributed minimally to whole-community decomposition during the winter, but during the summer dominated decomposition in the two arid sites. Using pitfall trapping, we found that macro-decomposer richness, abundance and biomass followed a hump-shaped response to precipitation, peaking in arid environments.

The puzzle of why plant litter decomposition in arid-lands is decoupled from annual precipitation and is faster than expected based on microbial decomposition models has bothered scientists for half a century<sup>7,34,35</sup>, and was later termed the desert decomposition conundrum<sup>36</sup>. Attempts to resolve this conundrum have focused predominantly on abiotic weathering agents, such as photodegradation<sup>37,38</sup> and thermal degradation<sup>39</sup>, alternative sources of moisture such as fog, dew and atmospheric water vapor<sup>40</sup> and soil-litter mixing<sup>36,41</sup>. We, in turn, hypothesized that the opposing climatic dependencies of macrofauna and that of microorganisms and mesofauna should lead to similar overall decomposition rates across precipitation gradients, except in hyperarid environments in which decomposers activity is predicted to be minimal regardless of organism size. Our results largely agree with this hypothesis. Whole-community decomposition was minimal in hyper-arid sites in both summer and winter. In the winter, microbial decomposition dominated the whole-community decomposition, demonstrating a positive response to precipitation that reached a maximum in the most mesic Mediterranean site. In contrast, macro-decomposition has contributed only little to the whole-community decomposition during the winter, but dominated the arid sites' decomposition in the summer. These findings supported the long-suggested but largely overlooked hypothesis that macro-decomposition governs plant litter decomposition in deserts<sup>7,34,35</sup>. The opposing climatic dependencies of micro- and macro-decomposers have led to similar or even higher annual decomposition rates in arid sites compared to those measured in more mesic sites. Consequently, we highlighted that differential climatic dependencies of different-sized decomposers rather than abiotic factors explain the discrepancy between classic decomposition models and the observed decomposition rates in drylands. This realization provided a plausible resolution to the longstanding desert decomposition conundrum, and exposed a hidden mechanism that may account for unexplained variation in plant litter decomposition across biomes.

Canonically, faunal decomposition is expected to be positively associated with temperature and moisture<sup>20</sup>. We, however, hypothesized that climate dependencies of mesofauna and macrofauna should differ due to the lower sensitivity of macrofauna to high temperature and low moisture, and the ability of macro-decomposers to shuttle between the hostile environment aboveground and the climatic havens belowground<sup>28</sup>. We also hypothesized that low and unpredictable resource availability in hyper-arid environments should limit macro-decomposer populations. Consequently, we predicted that macro-decomposers should be more prevalent in arid environments in comparison to hyper-arid or more mesic environments. Furthermore, ample resource availability may increase niche space<sup>42</sup>, resulting in higher macro-decomposer diversity, which in turn can facilitate decomposition through synergistic effects of functionally complementary species<sup>43–46</sup>. Thus, we predicted that macro-decomposition should reflect the variation in the abundance, richness and biomass of macro-decomposers across the precipitation gradient. Our findings supported these predictions. The richness, abundance and biomass of macro-decomposers followed a hump-shaped relationship with precipitation, peaking in arid environments and diminishing toward hyper-arid or semiarid and Mediterranean sites. Macro-decomposer assemblages were dominated by ants and beetles across the aridity gradient except in the Mediterranean site that was dominated by isopods and millipedes. During the summer, the

observed hump-shaped relationship between macro-decomposition and precipitation tightly echoed the variation in richness, abundance, and biomass of macro-decomposers, revealing the mechanistic foundation for the cross-system variation in macro-decomposition.

In winter, macro-decomposer abundance, richness, and biomass were similar to or even higher than those measured during the summer across all sites. Despite this resemblance, macro-decomposition did not reflect the observed variation in macro-decomposer assemblages. This discrepancy could be explained by between-seasons differences in the structure of the macro-decomposer assemblages (Fig. 6B). The macro-decomposer summer assemblage in Sayeret Shaked was more similar to the Avdat summer assemblage than to the Sayeret Shaked winter assemblage. The Avdat assemblages were more similar to each other across seasons than the Sayeret Shaked assemblages. This may explain why macro-decomposition in winter was higher in Avdat than in Sayeret Shaked. Termites (*Hodotermitidae* sp.), that play an important role in decomposition, were more abundant in summer compared to the winter in Sayeret Shaked but not in Avdat. Moreover, our data revealed that several beetle taxa (*Adelostoma* sp., *Akis reflexa* (Fabricius, 1775), *Dailognatha crenata* (Reiche & Saulcy, 1857), *Tentyrina* sp., *Zophosis* sp.) were prevalent in both arid sites during the summer but were absent or very scarce in the winter. Phenological differences in the behavior of dominant macro-decomposers may also contribute to the seasonal differences<sup>47</sup>. For instance, *Hemilepistus reaumuri* (Milne Edwards, 1840), an abundant isopod species in Avdat, and *Anacanthotermes ubachi* (Navás, 1911), a common termite species in Sayeret Shaked, consume detritus predominantly during the summer and autumn and disperse and reproduce during the winter<sup>48,49</sup>. Future studies should explicitly test these explanations. Regardless, the whole-year association between macro-decomposition and the abundance, richness, and biomass of macro-decomposers strongly support our hypothesis.

Theory suggests that plant litter decomposition by meso-decomposers should increase with moisture. This pattern was supported by a cross-biome experiment<sup>50</sup>. Thus, we hypothesized that both meso-decomposition and microbial decomposition should increase with precipitation and be more prominent in the winter than in the summer. Our results did not coincide with these hypotheses. Litter decomposition by mesofauna followed a unimodal pattern across the precipitation gradient, peaking under semiarid conditions in both seasons (Fig. 4). Meso-decomposition and macro-decomposition were similar in the hyper-arid and Mediterranean sites. However, meso-decomposition was higher than macro-decomposition in the semiarid sites and much lower than macro-decomposition in the arid sites. These results suggest that meso-decomposers, like macro-decomposers, have adaptations that allow them to thrive in moisture-deprived environments. Yet, meso-decomposition peaked in more mesic conditions than macro-decomposition, implying higher moisture dependency. Our findings highlight the need to explore the moisture dependency of mesofauna in greater details, and to generate different predictions for how mesofauna and microorganisms may affect plant litter decomposition.

Faunal decomposition in our study peaked in arid environments, contrasting the positive association between faunal decomposition and precipitation that was found in recent global meta-analyses<sup>20,21</sup>. This discrepancy may reflect underestimation of faunal decomposition rates in drylands, possibly because these studies either deliberately grouped cold and dry environments together<sup>20,50</sup>, or focused solely on precipitation without accounting for differences in temperature<sup>21</sup>. In cold water-deprived environments and seasons, low temperatures may limit the populations and activity of ectotherm animals<sup>4,22</sup>. Therefore, ignoring the effect of temperature may lead to falsely smaller faunal effects on decomposition in drylands. This bias may contribute to the positive association between precipitation and faunal decomposition. To reveal the realistic relationships, future studies on faunal decomposition should explore the effects of temperature, precipitation and the interaction between them. It is important noting that temperature could affect decomposition both directly by determining the activity of ectotherms and indirectly by regulating moisture availability. Thus, using aridity indices that aim to correct for moisture availability cannot resolve the need to account also for temperature *per se*.



We used litter of one plant species that is common to all seven experimental sites. In this way, we were able to compare decomposition rates across sites without using litter that is exogenous to one or more sites. Decomposer communities are assumed to perceive litter quality based on the chemical composition of the focal litter, and that of past resource inputs that these communities have experienced<sup>51</sup>[↗](#). Plant communities differ substantially between our sites, suggesting that each decomposer community has experienced very different resource inputs, and may have access to alternative litter of various qualities. Thus, it is likely that the perceived quality of *S. capensis* litter differs between sites, possibly confounding our results. Moreover, our sites were distributed along a single precipitation gradient. Thus, our results may be confounded by other environmental factors that were not accounted for. To determine how general our findings are, future research should repeat our experimental approach using different litter sources, and across multiple climatic gradients.

In conclusion, our work revealed that decomposers of varying size categories have different moisture dependencies. This suggests that microorganisms, meso-decomposers and macro-decomposers should be considered separately in decomposition models, and emphasizes the need to contemplate animals' physiology and behavior when investigating zoogeochemical processes. Warm drylands cover 19% of the land surface worldwide and expand rapidly due to unsustainable land-use and climate change<sup>52</sup>[↗](#). We highlight the importance of macro-decomposition in arid-lands that compensates for the minimal microbial decomposition, providing a plausible resolution to the long-debated dryland decomposition conundrum. Understanding the mechanisms that regulate decomposition in drylands is key for conserving and restoring fundamental ecosystem processes in these ever-growing areas, and in improving our understanding of global processes like C cycling. To date, the general conceptualization of decomposition is largely based on ample research from temperate ecosystems. Thus, prevailing theory centers on focal processes that dominate decomposition in these systems. Our work highlights that in other less studied ecosystems additional processes like the role of animal decomposers may be dominant, opening the door for new exciting research that may shake our conceptualization of decomposition processes.

## Authors' contributions

D.H., N.S. and V.T. conceived this study and designed the experiments. V.T, N.S. and D.H performed the field experiment. N.S. and V.T. performed lab analyses. J.A.D., Y.H. and E.G.R. sorted and identified animal samples. N.S. analyzed the data. N.S., V.T and D.H wrote the manuscript, and all authors provided critical feedback and approved submission.

## Data sharing plans

All data used in the manuscript has been deposited in an open repository and can be found on the following link <https://figshare.com/s/96ff438c98ae4a6afbc4> [↗](#)

## Acknowledgements

We thank Liat Hadar, Ronen Kadmon, Ronen Ron, Omri Sharon, Shayli Dor-Haim, Nitzan Segev and Rachel Armoza-Zvuloni for facilitating our work in the experimental sites; Igor Armiach Steinpress, Laibale Friedman and Armin Ionescu for help with morphospecies identification; and Ariel Malinsky-Buller, Netanel Paz, Omri Sherman, Dor Gabay, Coral Ben-Lulu, Ofer Frumkin, Aparna Lajmi, and Atar, Alon and Ela Hawlena for lab and field assistance.

## Funding

The research was funded by the Israel Science Foundation (ISF-No. 1391/19) to D.H.

## Competing interests

The authors declare no conflict of interest.

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## Editors

Reviewing Editor

**David Donoso**

National Polytechnic School, Quito, Ecuador

Senior Editor

**Meredith Schuman**

University of Zurich, Zürich, Switzerland

## Reviewer #1 (Public Review):

Summary:

Torsekar et al. use a leaf litter decomposition experiment across seasons, and in an aridity gradient, to provide a careful test of the role of different-sized soil invertebrates in shaping the rates of leaf litter decomposition. The authors found that large-sized invertebrates are more active in the summer and small-sized invertebrates in the winter. The summed effects of all invets then translated into similar levels of decomposition across seasons. The system breaks down in hyper-arid sites.

<https://doi.org/10.7554/eLife.93656.2.sa1>

## Author response:

The following is the authors' response to the original reviews.

### **Public Reviews:**

#### **Reviewer #1 (Public Review):**

Summary:

*I really enjoyed this manuscript from Torsekar et al on "Contrasting responses to aridity by*

*different-sized decomposers cause similar decomposition rates across a precipitation gradient". The authors aimed to examine how climate interacts with decomposers of different size categories to influence litter decomposition. They proposed a new hypothesis: "The opposing climatic dependencies of macrofauna and that of microorganisms and mesofauna should lead to similar overall decomposition rates across precipitation gradients".*

*This study emphasizes the importance as well as the contribution of different groups of organisms (micro, meso, macro, and whole community) across different seasons (summer with the following characteristics: hot with no precipitation, and winter with the following characteristics: cooler and wetter winter) along a precipitation gradient. The*



authors made use of 1050 litter baskets with different mesh sizes to capture decomposers contribution. They proposed a new hypothesis that was aiming to understand the "dryland decomposition conundrum". They combined their decomposition experiment with the sampling of decomposers by using pitfall traps across both experiment seasons. This study was carried out in Israel and based on a single litter species that is native to all seven sites. The authors found that microorganism contribution dominated in winter while macrofauna decomposition dominated the overall decomposition in summer. These seasonality differences combined with the differences in different decomposers groups fluctuation along precipitation resulted in similar overall decomposition rates across sites. I believe this manuscript has a potential to advance our knowledge on litter decomposition.

**Strengths:**

Well design study with combination of different approaches (methods) and consideration of seasonality to generalize pattern.

The study expands to current understanding of litter decomposition and interaction between factors affecting the process (here climate and decomposers).

**Weaknesses:**

The study was only based on a single litter species.

We now discuss the advantages and limitations of this approach in the methods and devote a completely new paragraph to this important point in the discussion (lines 394-401).

**Reviewer #2 (Public Review):**

*Summary:* Torsekar et al. use a leaf litter decomposition experiment across seasons, and in an aridity gradient, to provide a careful test of the role of different-sized soil invertebrates in shaping the rates of leaf litter decomposition. The authors found that large-sized invertebrates are more active in the summer and small-sized invertebrates in the winter. The summed effects of all invets then translated into similar levels of decomposition across seasons. The system breaks down in hyper-arid sites.

*Strengths:* This is a well-written manuscript that provides a complete statistical analysis of a nice dataset. The authors provide a complete discussion of their results in the current literature.

**Weaknesses:**

I have only three minor comments. Please standardize the color across ALL figures (use the same color always for the same thing, and be friendly to color-blind people).

Thank you for this important suggestion. We have now changed all figures to standardize all colors and chose a more color-blind friendly palette.

Fig 1 may benefit from separating the orange line (micro and meso) into two lines that reflect your experimental setup and results. I would mention the dryland decomposition conundrum earlier in the Introduction.

We based our novel hypotheses on a thorough literature search. Accordingly, decomposition is expected to be positively associated with moisture, regardless of the decomposer body size. Our contribution to theory was to suggest that macro-detritivores may respond very differently to climatic conditions and dominate litter decomposition in warm arid-lands (we

listed the reasons in the text). Consequently, we did not distinguish between microorganisms and mesofauna. We assumed that both groups inhabit the litter substrate and have limited adaptation to dry conditions. Our results provide strong evidence that this presumption is likely wrong and that mesofauna respond to climate very differently from micro-decomposers. Yet, we cannot use hindsight understanding to improve our original hypothesis. We now emphasize this important point at the discussion as important future direction.

Although we are very appreciative and pleased with the reviewer enthusiasm to highlight the importance of our work as a possible solution to the longstanding dryland decomposition conundrum, we decided not to move it to the introduction. This is because we think that our work is not centred on resolving the DDC but provides more general principles that may lead to a paradigm shift in the way ecologists study nutrient cycling across ecosystems.

*And the manuscript is full of minor grammatical errors. Some careful reading and fixing of all these minor mistakes here and there would be needed.*

We apologize and did our best to find and fix those mistakes

### **Recommendations for the authors:**

#### **Reviewer #1 (Recommendations For The Authors):**

*I really enjoyed this manuscript from Torsekar et al on "Contrasting responses to aridity by different-sized decomposers cause similar decomposition rates across a precipitation gradient". The authors aimed to examine how climate interacts with decomposers of different size categories to influence litter decomposition. They proposed a new hypothesis: "The opposing climatic dependencies of macrofauna and that of microorganisms and mesofauna should lead to similar overall decomposition rates across precipitation gradients".*

*This study emphasizes the importance as well as the contribution of different groups of organisms (micro, meso, macro, and whole community) across different seasons (summer with the following characteristics: hot with no precipitation, and winter with the following characteristics: cooler and wetter winter) along a precipitation gradient. The authors made use of 1050 litter baskets with different mesh sizes to capture decomposers contribution. They proposed a new hypothesis that was aiming to understand the "dryland decomposition conundrum". They combined their decomposition experiment with the sampling of decomposers by using pitfall traps across both experiment seasons. This study was carried out in Israel and based on a single litter species that is native to all seven sites. The authors found that microorganism contribution dominated in winter while macrofauna decomposition dominated the overall decomposition in summer. These seasonality differences combined with the differences in different decomposers groups fluctuation along precipitation resulted in similar overall decomposition rates across sites.*

*I believe this manuscript has the potential to advance our knowledge on litter decomposition. Below i provide my general and specific comments.*

*General comments:*

- (1) Study in general is well designed and well thought beforehand,*
- (2) Study aims to expand the current understanding of the dryland decomposition conundrum*
- (3) The should put a caveat to the fact they only use one litter species and call for examining litter mixture in the same gradient.*

*(4) Please check the way you reduce the random effects from your initial model, I have provided a better way to do so in my specific comments*

*(5) For Figure 1, authors can check my comment on this and see if they could revise the figure.*

Thank you for the positive feedback and your valuable comments. We have tried to best address all comments and suggestions for improvement and clarification

*Specific comments*

*Line # 57 Please write "Theory suggests" instead of "Theory suggest"*

We changed the text as suggested

*Line # 70, please write "Indeed, handful evidence shows" instead of "Indeed, handful evidence show"*

We changed the text as suggested

*Figure 1: I like this conceptual framework. I have a silly question, why is it that the slopes of the whole community at the beginning (between Hyperarid and Arid) is the same as the Macro fauna, I would think the slope should be higher as this is adding up right? and also the same goes for the decomposition of whole community later on. For me this should reflect the adding or summing up (if i am right) then the authors should think about how this could be reflected in the figure.*

We agree with your interpretation that the whole community decomposition reflects the addition by constituent decomposers. The slope of the whole community decomposition between hyper-arid and arid is slightly higher than the one of macro decomposition to reflect the additive effect of macro with meso+micro decomposition. We have now changed the figure slightly to make this point more visible (Line 106).

*Line # 111 Please make "Methods" bold as well to be consistent with others headings.*

We changed the formatting as suggested

*Line #125 and in other lines as well please replace "X" by "x" to denote multiplication.*

We changed the formatting as suggested

*Table 1 Please add "\*" to climate like this "Climate\*" so that the end note of the table could make sense*

Thank you for this suggestion. We have now added the asterisk referring to the note below the Table.

*Figure 2, please consider putting at line #133, mean annual precipitation (MAP), as such for line # 135 You can directly says The precipitation map ....*

We made both changes as suggested.

*Line # 138 I would not use the different units for the same values. I do understand that you want to emphasize the accuracy but i would write instead  $3 \pm 0.001$  g*

We changed the units as suggested.

*Line # 145, how is the litter basket customized to rest at 1 cm above ground level?*

We have now clarified –that we cut-open windows one centimeter above the cage floor. The cages were positioned on the soil (line 144).

*Lines # 181-183, I like the approach of checking the necessity of having the random effects. However, it has been reported that likelihood ratio test (LRT) are not really reliable to test for random effects. I will suggest you rather use permutations instead. I think the function is confint(MODEL) you need to specify the number of permutation the higher the better but you should start with 99 first and see how the results look like if promising then you can even go to 9999. But it will need computation power and and time.*

Thank you for the suggestion. We now used a simulation-based exact test, instead of a LRT, to examine the random effect, as recommended by the authors from the “lme4” package. As recommended, we used 9999 simulations. The simulation test yielded a similar result to those originally reported (see lines 181-183).

*Line # 187, 188, 188, please do not use capital letter to start mesofauna, macrofauna and whole-community*

We changed the formatting as suggested

*Line # 205 Please add the version number of R in the text.*

We now included the version number as suggested.

*Line # 209-211, could you please check whether "then" is the word you want to use or "than"*

Our bad- we indeed meant “than” and have made the appropriate changes.

*Line # 227 and in other places as well please provide the second degree of freedom of the F test.*

Thank you for this important comment. We have now added the second degree of freedom to the relevant results (lines 229, 232).

*Figure 3 and Figure 4 show some results that are negative, can you please explain what might be the reasons behind this?*

We now explain this important point in the figures’ captions.

*Figure 5 Please add label to the x-axis.*

Thank you-we have now included a label.

*Line # 357, the sentence "... meso-decomposition, like microbial decomposition,...", I don't understand which criteria authors used to classify microbial decomposition as "meso-decomposition"?*

We now remove this potential cause of confusion by using the term ‘meso-decomposition’ to distinguish from microbial decomposition (Line 366).

| *Line # 380 Kindly put "per se" in italic.*

We changed the formatting as suggested

*References*

*The references format are not consistent. For example for the same journal (say Trends in Ecology and Evolution) the authors sometimes wrote the full name like at line # 36 (and also realize that "vol" should not be written as such) but wrote the abbreviations at line #42*

Our bad- we apologize and carefully checked all references to make sure the style is consistent.

<https://doi.org/10.7554/eLife.93656.2.sa0>